

Table 31: Evaluation results across all experimental settings, reported as macro-averaged Relative Mapping Similarity (RMS) precision, recall, and F_1 (in %). Setting 1 is the symbolic SPARQL baseline over ORKG comparisons. Setting 2 comprises neural querying over PDFs (2.1: proprietary models; 2.2a: open-weights with a single concatenated PDF context; 2.2b: open-weights with RAG over embedded PDF chunks). Setting 3 comprises symbolic-context-augmented neural QA over ORKG comparison CSVs (3.1: proprietary models; 3.2: open-weights). The “#queries” column indicates, for each system, how many of the 33 ALD+ALE queries produced a valid prediction that could be evaluated against the SPARQL gold-standard tables. Open-weights shorthand names: “gemma-3” = gemma-3-27b-it; “llama-3.3” = Llama-3.3-70B-Instruct; “mistral” = Mistral-Large-Instruct-2407; “qwen2.5-vl” = Qwen2.5-VL-72B-Instruct; “qwen3” = Qwen3-30B-A3B-Instruct-2507.

Setting	Model type	Context	Model	#queries	RMS-prec. [%]	RMS-rec. [%]	RMS-F1 [%]
1	Symbolic	ORKG comparison (SPARQL over graph)	SPARQL	33	100.0	100.0	100.0
2.1	Prop.	Review article PDF (black-box proprietary PDF processing)	chatgpt-5.1	33	46.0	39.5	40.6
			claude-sonnet	33	61.4	63.7	61.7
			gemini-3	33	59.6	60.0	59.1
2.2a	Open	PDF text and tables (single concatenated context)	gemma-3	33	25.5	25.1	24.6
			llama-3.3	33	32.6	35.9	32.8
			mistral	32	28.4	27.1	25.9
			qwen2.5-vl	33	34.1	27.6	28.2
			qwen3	32	29.1	33.9	29.7
2.2b	Open	PDF text and tables (RAG over embedded chunks)	gemma-3	33	21.4	18.9	19.0
			llama-3.3	33	25.0	23.8	22.8
			mistral	33	23.8	20.9	21.1
			qwen2.5-vl	33	28.2	24.1	24.3
			qwen3	33	24.7	24.6	23.2
3.1	Prop.	ORKG comparison exported as CSV (symbolic table context)	chatgpt-5.1	33	67.8	61.5	63.0
			claude-sonnet	33	74.0	76.1	74.2
			gemini-3	33	72.2	71.5	70.3
3.2	Open	ORKG comparison exported as CSV (symbolic table context)	gemma-3	33	44.8	47.4	44.7
			llama-3.3	33	48.5	52.8	49.3
			mistral	31	51.4	51.2	48.8
			qwen2.5-vl	32	46.0	37.9	39.3
			qwen3	32	40.8	46.0	40.8

4.2.2 Neural querying over PDFs: performance on precise scientific Q&A tasks

Here we address RQ2, i.e., how well large language models that only see the review-article PDFs can reproduce the precise, table-style answers obtained from symbolic SPARQL querying. The corresponding Relative Mapping Similarity (RMS) scores (introduced in subsection 4.1.3) are summarised in Table 31 for Settings 2.1, 2.2a, and 2.2b. In Setting 2.1, we use the chat interfaces of three proprietary models (ChatGPT-5.1, Gemini-3-Pro, and Claude-Sonnet-4.5), uploading the full PDF and treating their internal PDF handling as a black box. In Settings 2.2a and 2.2b, we evaluate open-weights models using a transparent pipeline: text and tables are first extracted from each PDF and then provided as input, either as one long context (2.2a) or via a simple retrieval step over document segments (2.2b). Implementation details are in the Appendix.

Within this purely PDF-based regime, the proprietary models in Setting 2.1 perform best. Claude-Sonnet-4.5 reaches the highest RMS F_1 at 61.7%, followed by Gemini-3-Pro at 59.1%, with ChatGPT-5.1 at 40.6%. Even this weakest proprietary model outperforms all open-weights systems in Settings 2.2a and 2.2b, indicating that the proprietary PDF-processing and document-structure handling currently offer a substantial advantage over our simple open-source pipeline. Among the open-weights models with a single concatenated context (Setting 2.2a), the best performer, Llama-3.3-70B-Instruct, attains 32.8% RMS F_1 , with the remaining models clustered around 24–30%. Moving to the retrieval-augmented variant (Setting 2.2b) consistently *decreases* performance: all five open-weights models drop into the 19–24% RMS F_1 range, suggesting that our straightforward “retrieve a few segments and answer from them” strategy is not yet sufficient for reliable table-style question answering and can even omit important evidence.

Overall, these findings show that purely neural PDF-based querying, although workable to some extent, remains substantially below the symbolic SPARQL upper bound and is tightly constrained by the quality of the PDF-processing and context-selection pipeline. The proprietary systems currently set the bar in this regime, but even the best of them recovers only about 60% of the SPARQL RMS F_1 signal. For open-weights models, the gap widens further, and naïve retrieval does not close it. This

motivates the next subsection (RQ3), where we remove the PDF bottleneck and ground the same models directly in the machine-actionable ORKG comparison tables.

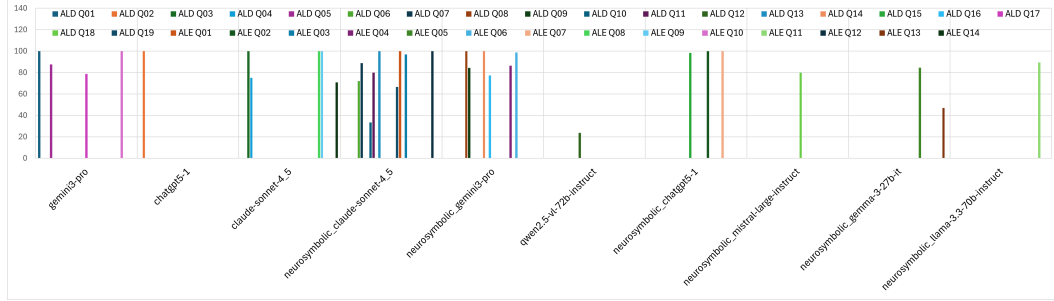
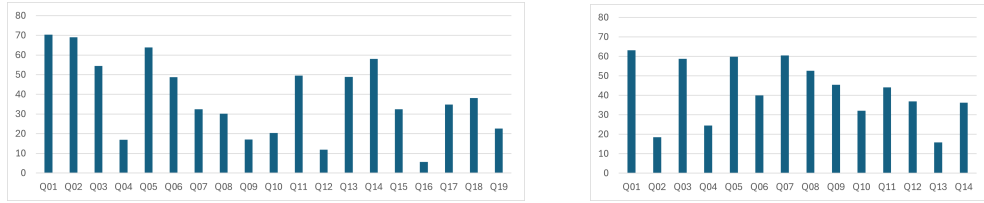
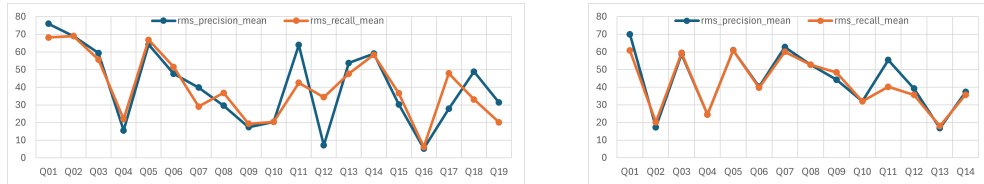


Figure 9: Best RMS F_1 score obtained for each query. For every ALD and ALE query, we identify the system with the highest RMS F_1 score when compared against the SPARQL gold-standard table. Among the 21 systems evaluated, 10 achieve the best RMS F_1 score on at least one query. Here, “best” denotes the highest RMS F_1 score attained for a given query across the 21 systems, not necessarily a perfect score of 100%.



(a) Mean RMS F_1 scores for the 19 ALD queries, averaged across 21 systems. (b) Mean RMS F_1 scores for the 14 ALE queries, averaged across the same 21 systems.

Figure 10: Per-query Relative Mapping Similarity (RMS) F_1 scores for ALD (left) and ALE (right). For each query, we compute RMS F_1 between each system’s predicted table and the SPARQL gold-standard table, then average the F_1 values over the 21 systems. Higher mean RMS F_1 scores indicate queries that are easier for the models to reproduce the symbolic SPARQL results, whereas lower scores indicate more challenging queries in the LLM setting.



(a) Mean RMS precision and recall for the 19 ALD queries, averaged across 21 neural and neurosymbolic queries, averaged across the same 21 systems. (b) Mean RMS precision and recall for the 14 ALE queries, averaged across the same 21 systems.

Figure 11: Per-query Relative Mapping Similarity (RMS) precision and recall for ALD (left) and ALE (right). For each query, we compute RMS precision and recall between each system’s predicted table and the SPARQL gold-standard table, then average these scores across the 21 systems. Points where the mean precision and recall curves coincide indicate that systems tend to reproduce the gold-standard mappings without adding or omitting entries. In contrast, a noticeable gap with substantially lower precision than recall indicates hallucination (many additional entries not supported by SPARQL), whereas higher precision than recall indicates under-coverage, i.e., models tend to produce only a conservative subset of the gold-standard entries.

4.2.3 Neural querying over ORKG comparisons: benefits of symbolic grounding

From an ALD/ALE perspective, RQ3 asks a simple question: if we give LLMs the *same kind of structured table* that experts use (the ORKG comparisons), do they get closer to the “ideal” SPARQL answers than when they have to read the PDFs by themselves? In contrast to Setting 2 (PDF-based querying), Setting 3 therefore uses the same proprietary and open-weights models but restricts their input to CSV-serialized ORKG comparison tables. Each model receives the exported comparison table together with the detailed natural-language query. This isolates neural reasoning over fully structured, machine-actionable evidence and lets us compare symbolic-context-augmented neural querying directly to the symbolic SPARQL gold standard (Setting 1) and to purely neural PDF-based querying (Setting 2). Relative Mapping Similarity (RMS) scores for all systems (introduced in subsection 4.1.3) are summarised in Table 31.

In Table 31, the last two main blocks (Settings 3.1 and 3.2) show a clear gain from symbolic grounding. With ORKG tables as context, all three proprietary models improve substantially over their PDF-only counterparts, and the best model now exceeds 70% RMS F_1 . Claude-Sonnet-4.5 reaches 74.2% RMS F_1 , Gemini-3-Pro 70.3%, and ChatGPT-5.1 63.0%. For an ALD/ALE reader, this means that when these systems are given a clean process table rather than a full PDF, roughly two thirds to three quarters of the rows and values in their answers match the SPARQL baseline. In the open-weights regime, Llama-3.3-70B-Instruct remains the strongest model and rises to 49.3% RMS F_1 in Setting 3.2, with Mistral-Large-Instruct-2407 close behind at 48.8%. Symbolic table context thus lifts performance for *all* model families and noticeably narrows (though does not close) the gap to the SPARQL upper bound of 100%.

Figure 9 shows, for each ALD and ALE query, which system achieves the highest RMS F_1 when compared to the SPARQL result. Most queries are “won” by systems operating in the symbolically grounded Setting 3, not by the PDF-only Settings 2.1–2.2. Put differently: whenever some model in our pool can solve a query reasonably well, it almost always does so when it sees the curated ORKG comparison table rather than the original PDF. This directly supports our central hypothesis that structured, machine-actionable evidence stores are more effective for precise information and knowledge synthesis than relying solely on the generative capabilities of LLMs over unstructured documents.

At the same time, symbolic grounding does not make all questions easy. Figure 10 shows, for each query, the average RMS F_1 across all 21 systems. Some questions reach relatively high average scores and can be answered reliably by many models once the ORKG tables are available. Others remain difficult even in Setting 3, with mean RMS F_1 sometimes below 20%. These harder cases tend to combine several ingredients: tight numeric constraints (e.g., simultaneous bounds on temperature, thickness, or etch rate), information scattered across long or multi-page tables, and the need to group or aggregate by higher-level categories (such as precursor families, linker types, or device classes) or to join information across multiple tables. For such multi-step reasoning tasks, small mistakes in filtering, grouping, or averaging quickly translate into missing or incorrect rows and are heavily penalised by the RMS metric.

Figure 11 adds a complementary view by plotting mean RMS precision and recall per query. When precision and recall are close, systems tend to reproduce the SPARQL mappings without adding many spurious entries or omitting many gold-standard rows. Noticeably lower precision than recall signals hallucination: models often output extra table entries that are not supported by the SPARQL results. Higher precision than recall indicates under-coverage, where systems return only a conservative subset of the true entries. Moving from PDFs (Setting 2) to ORKG tables (Setting 3) substantially reduces both hallucination and under-coverage, but does not eliminate them entirely, especially for the more complex queries described above.

Overall, RQ3 can be answered positively for ALD/ALE practice: when LLMs are grounded in structured ORKG comparison tables, they move significantly closer to the symbolic SPARQL baseline and behave in a more controlled way than when they work from PDFs alone. At the same time, the remaining errors point to clear avenues for improvement: richer and more complete comparison schemas, better handling of multi-constraint numeric filters and cross-table joins, and interaction patterns that guide models toward systematic filtering and aggregation rather than ad-hoc text generation. All natural-language queries, SPARQL formulations, answer tables, and ORKG comparisons are released as the *ALD/E Query Dataset* on Zenodo (DOI: <https://doi.org/10.5281/zenodo.17720429>) [15], enabling inspection, reuse, and extension by the ALD/ALE community.